

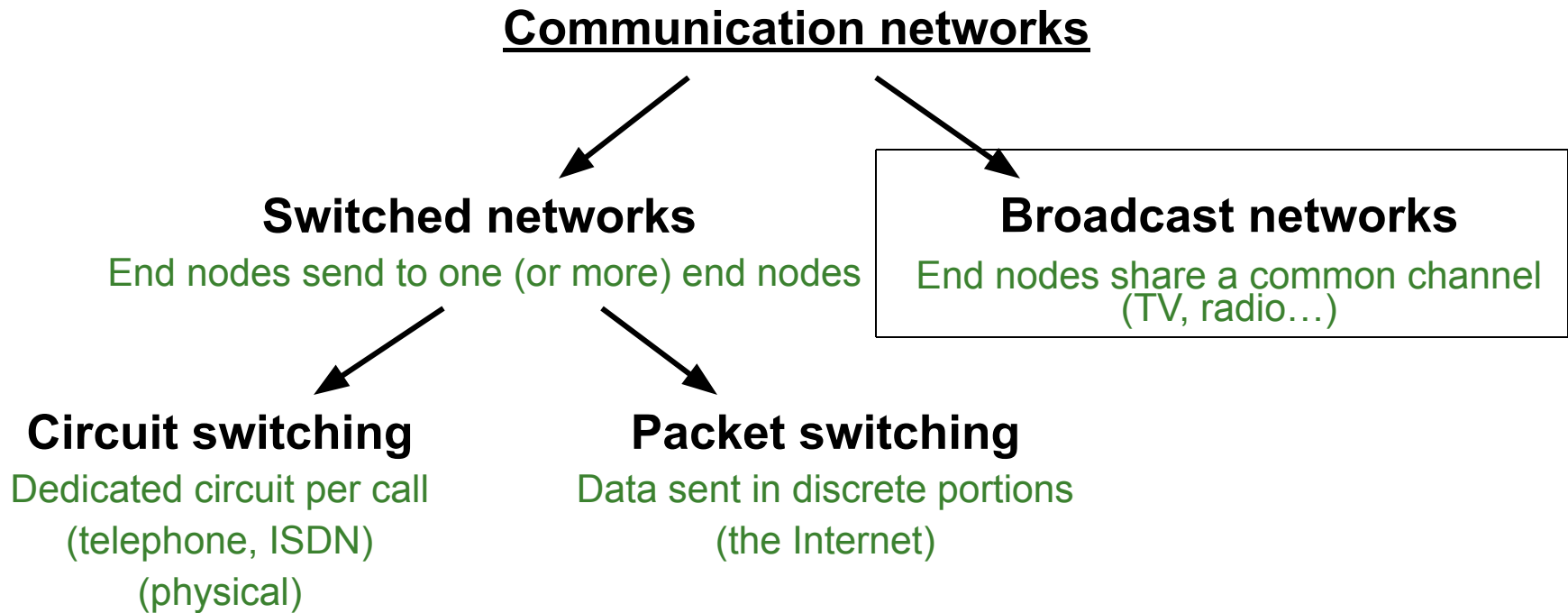
Computer Communication & Networks

Lecture 5 & 6

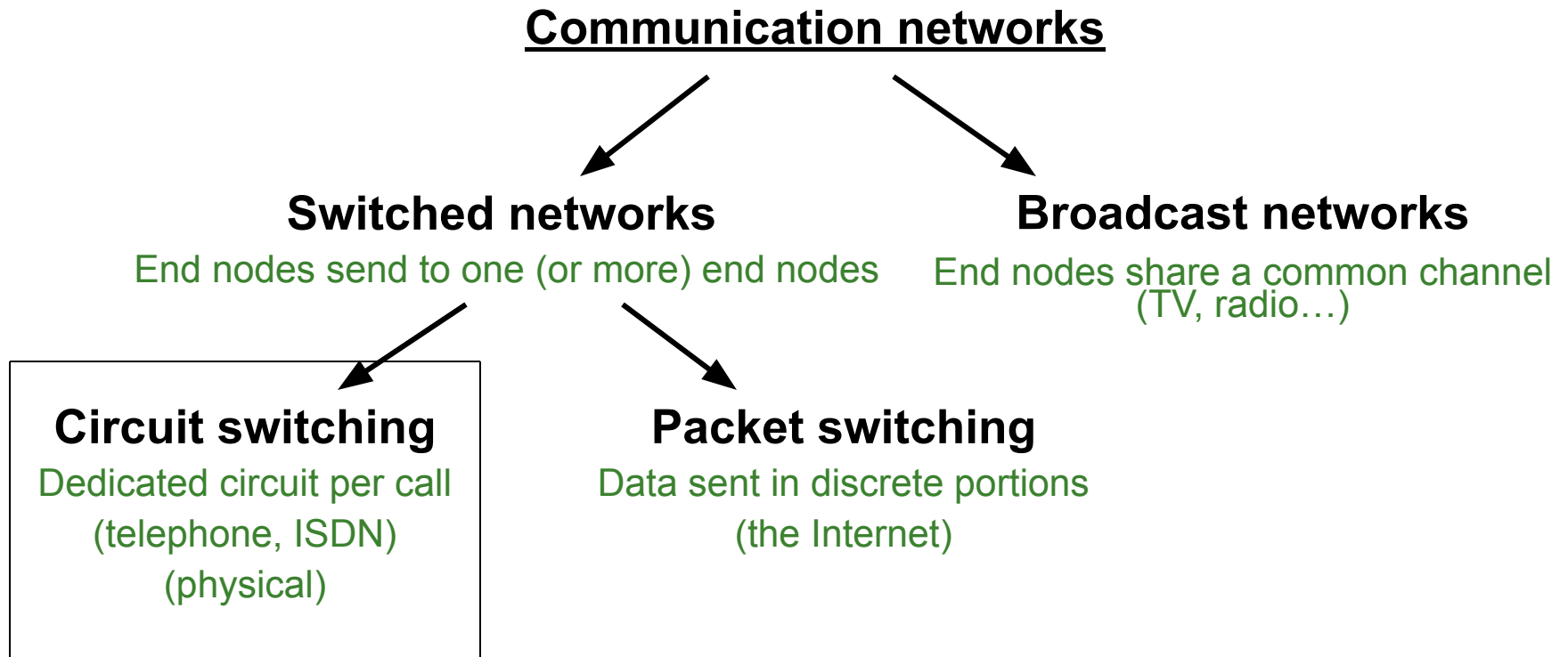
Circuit Switching, Packet Switching, Delays

Dr. Sobia Arshad

Communication Network



Communication Network



Circuit switching

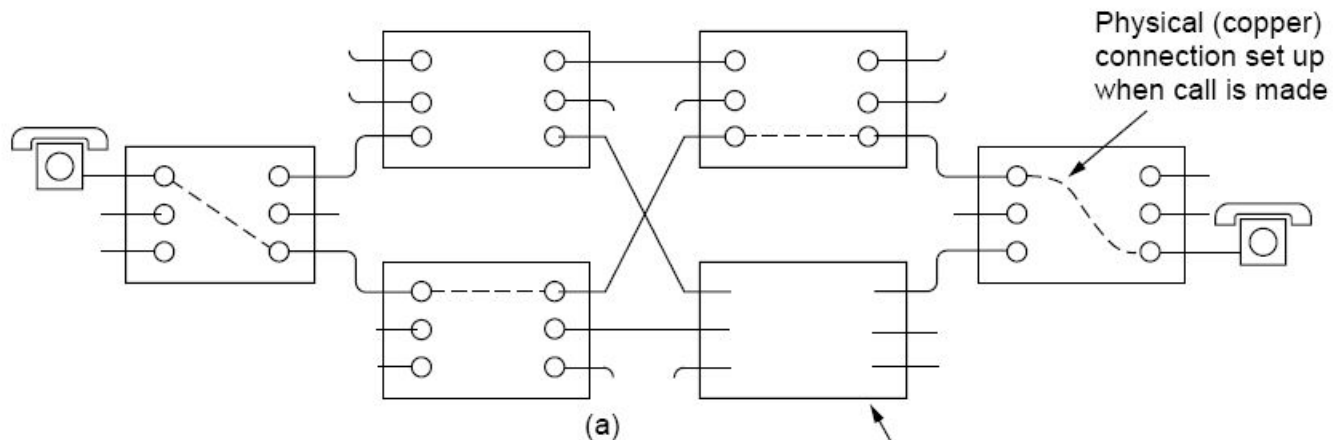
- A dedicated communication path (sequence of **links-circuit**) is established between the two end nodes through the nodes of the network
- Bandwidth: A circuit **occupies a fixed capacity** of each link for the entire lifetime of the connection. Capacity unused by the circuit cannot be used by other circuits.
- Latency: Data is **not delayed** at switches

Circuit switching (cnt'd)

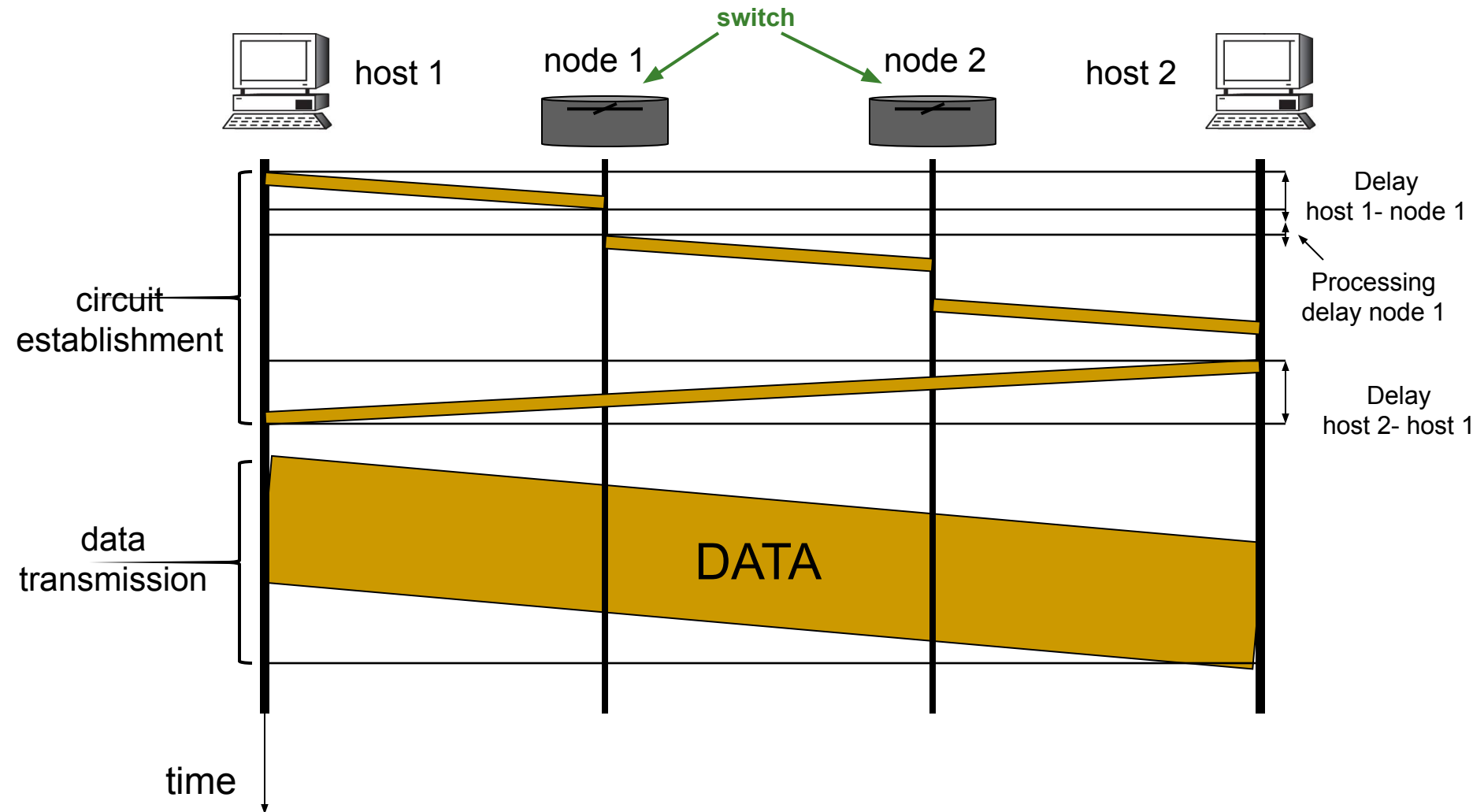
Three phases involved in the communication process:

1. Establish the circuit
2. Transmit data
3. Terminate the circuit

If circuit not available: busy signal (congestion)



Time diagram of circuit switching



Circuit Switching

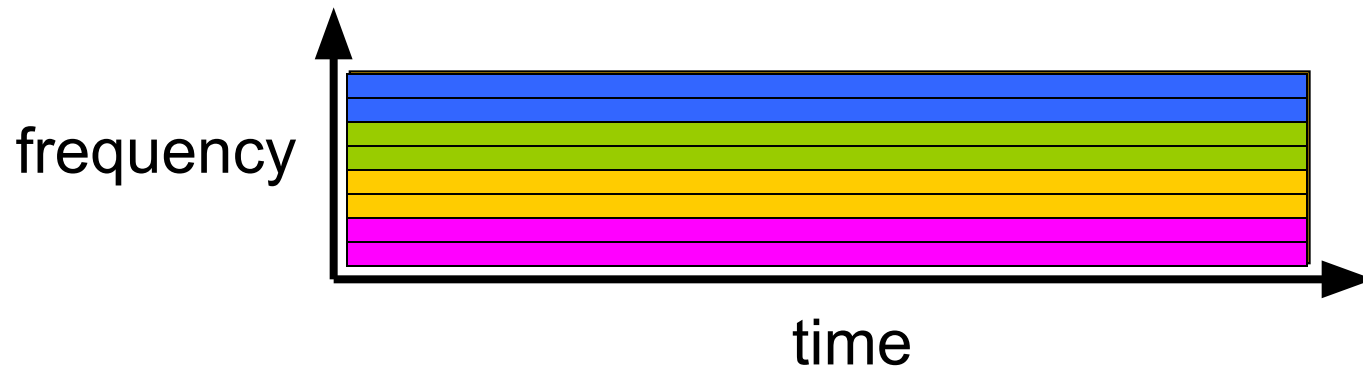
- Network resources (e.g., bandwidth) divided into “pieces”
- pieces allocated to calls
- resource piece **idle** if not used by owning call (*no sharing*)
- dividing link bandwidth into “pieces”
 - frequency division
 - time division

Circuit Switching: FDM and TDM

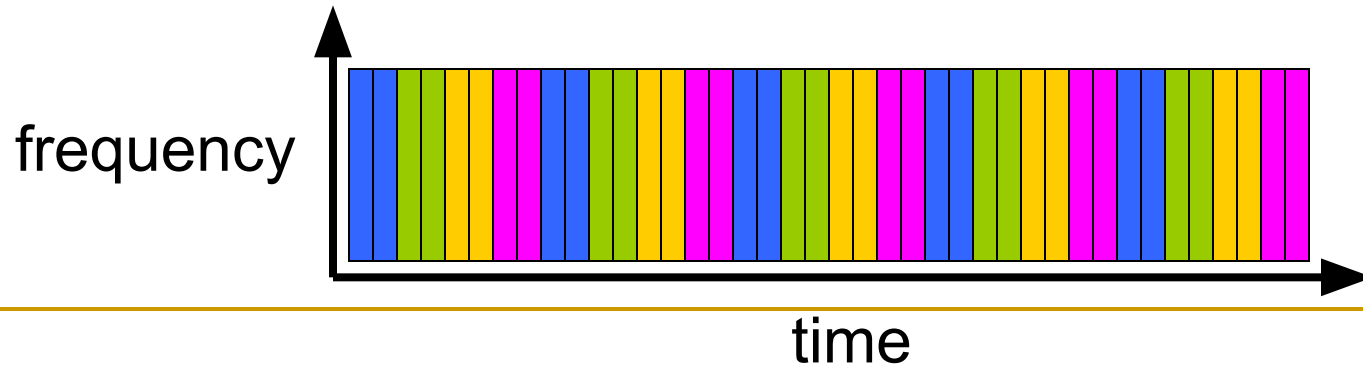
FDM

Example:

4 users



TDM



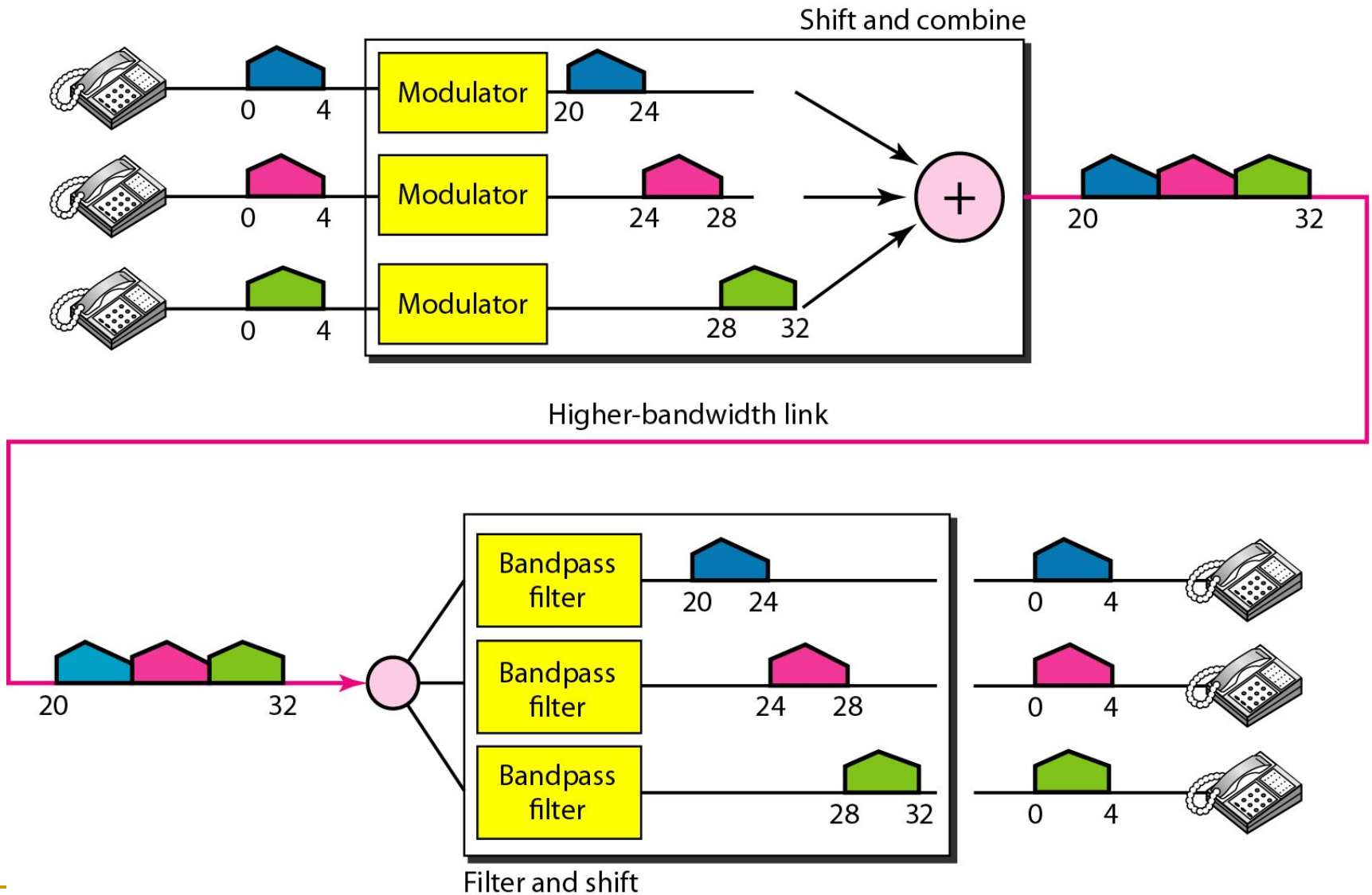
Example

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure on next Slide. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them.

Example (contd.)



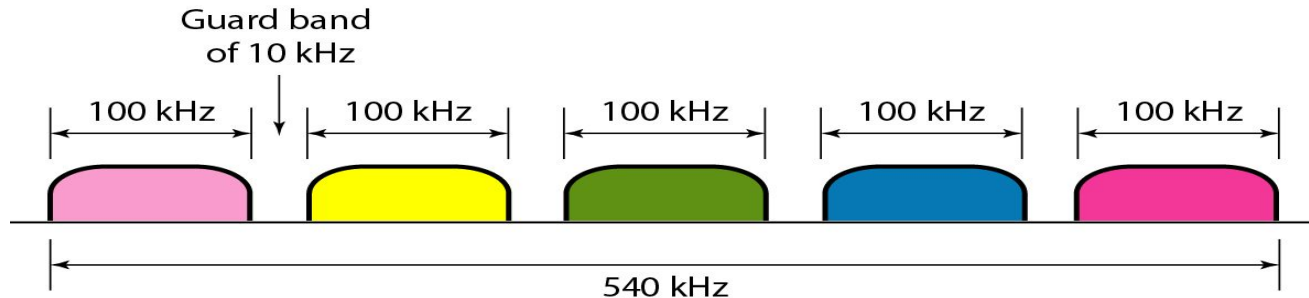
Example

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz}$$

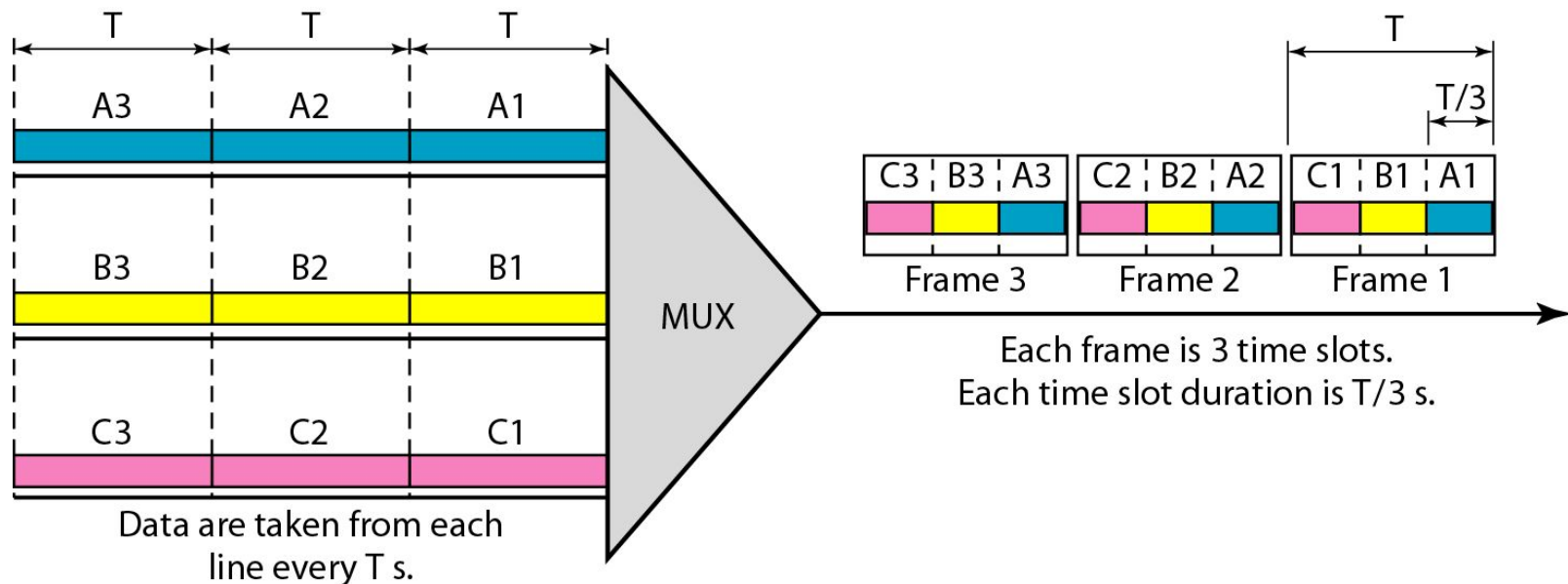


Applications

- AM Radio
 - Band 530-1700KHz
 - Each AM Station needs 10KHz
- FM Radio
 - Band 88-108MHz
 - Each FM Station needs 200KHz
- TV
 - Each Channel needs 6MHz
- AMPS

Synchronous TDM

- In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter.



Example

In Figure on Last Slide, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

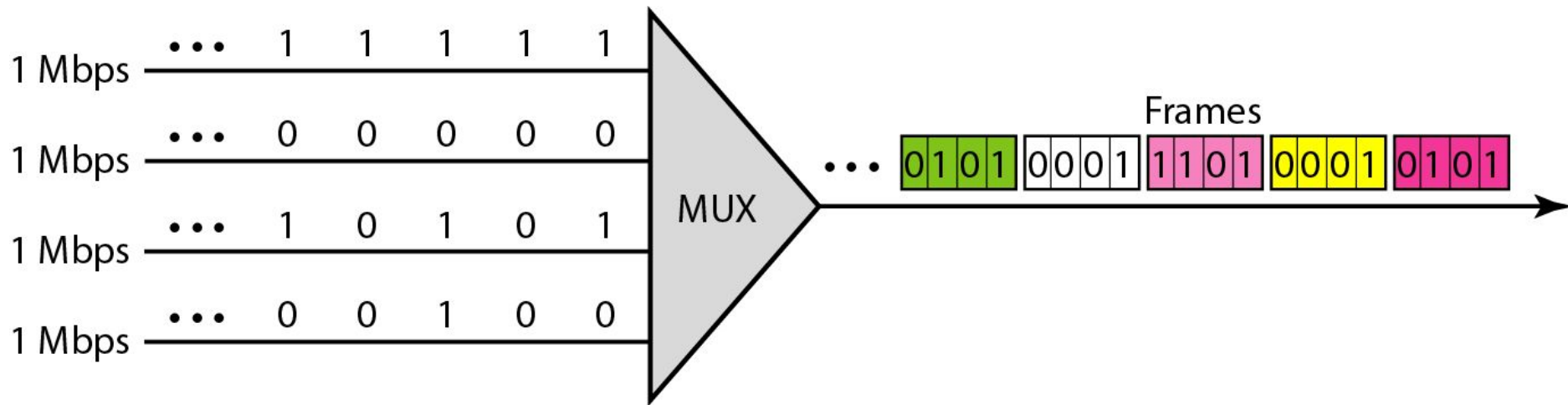
Solution

We can answer the questions as follows:

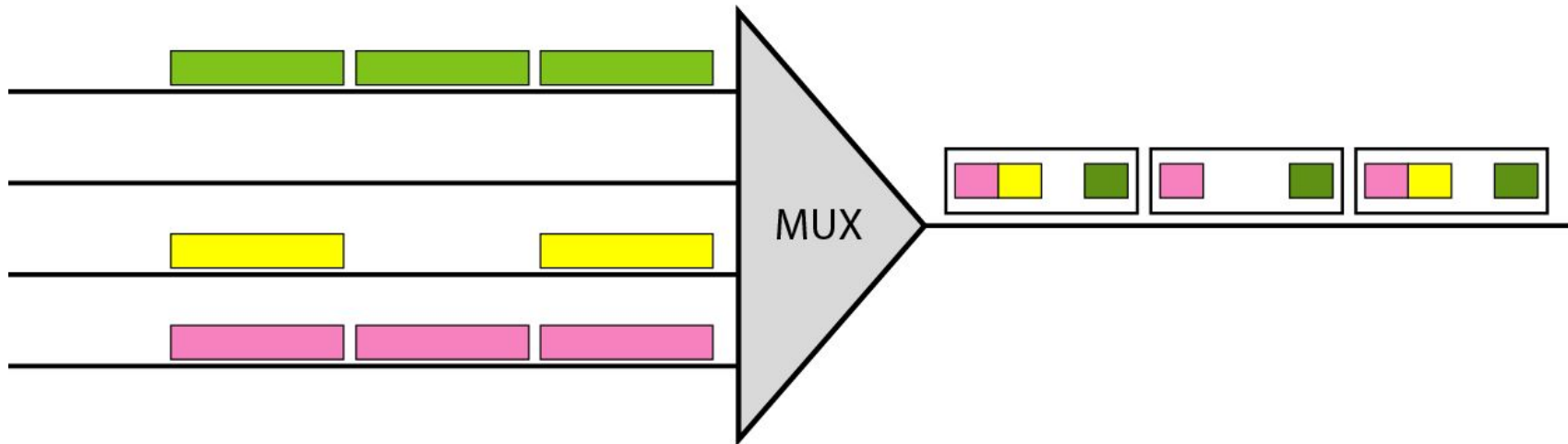
- a. The data rate of each input connection is 1 kbps. This means that the bit duration is $1/1000$ s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
- b. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is $1/3$ ms.
- c. Each frame carries three output time slots. So the duration of a frame is $3 \times 1/3$ ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.

Example

Figure below shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

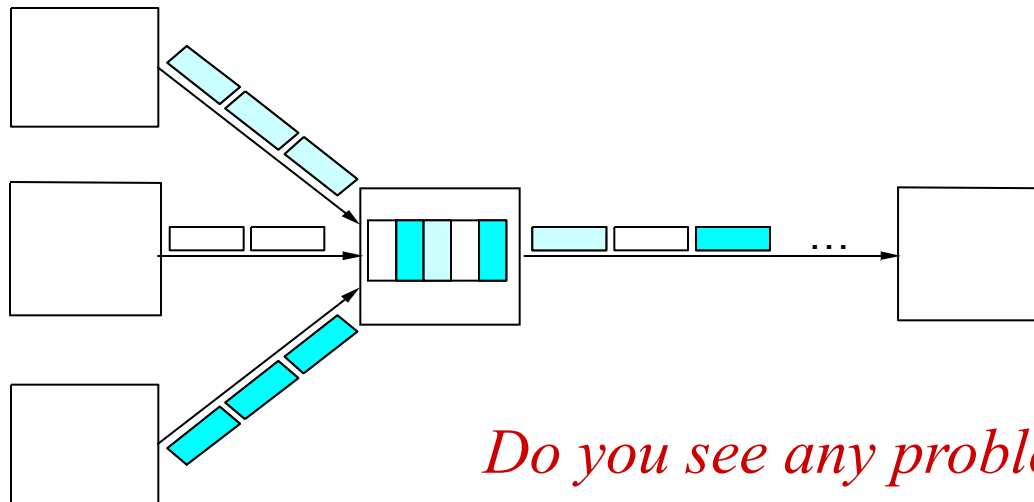


Disadvantages of Sync. TDM



Statistical Multiplexing

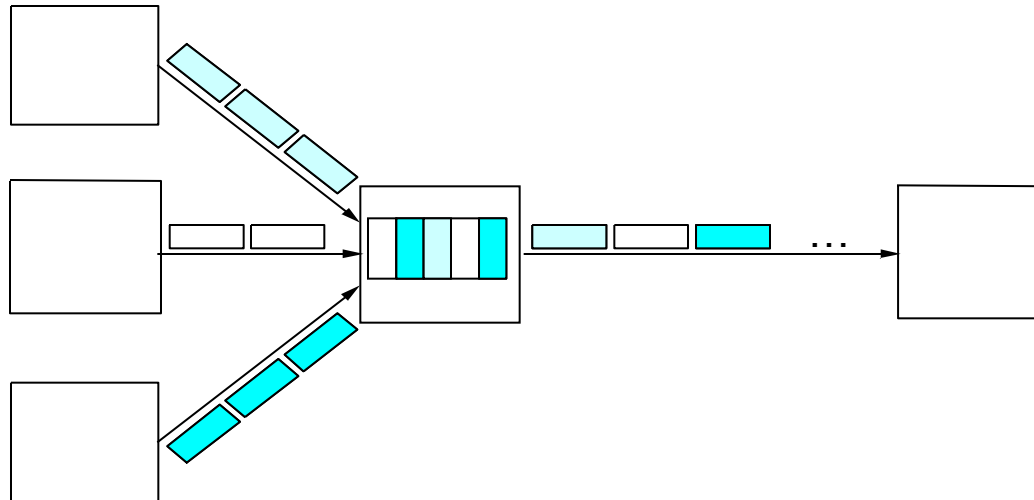
- On-demand time-division
- Schedule link on a per-packet basis
- Packets from different sources interleaved on link
- Buffer packets in switches that are *contending* for the link



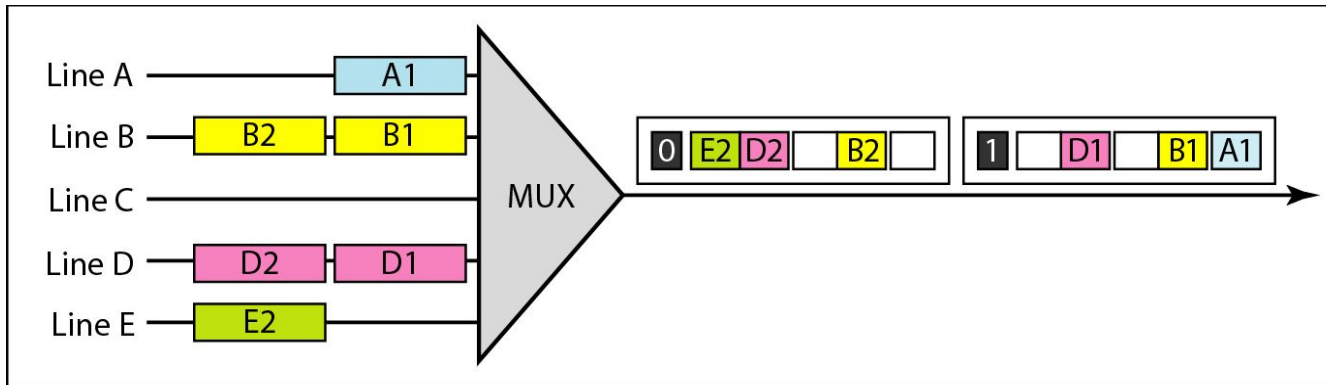
Do you see any problem ?

Statistical Multiplexing

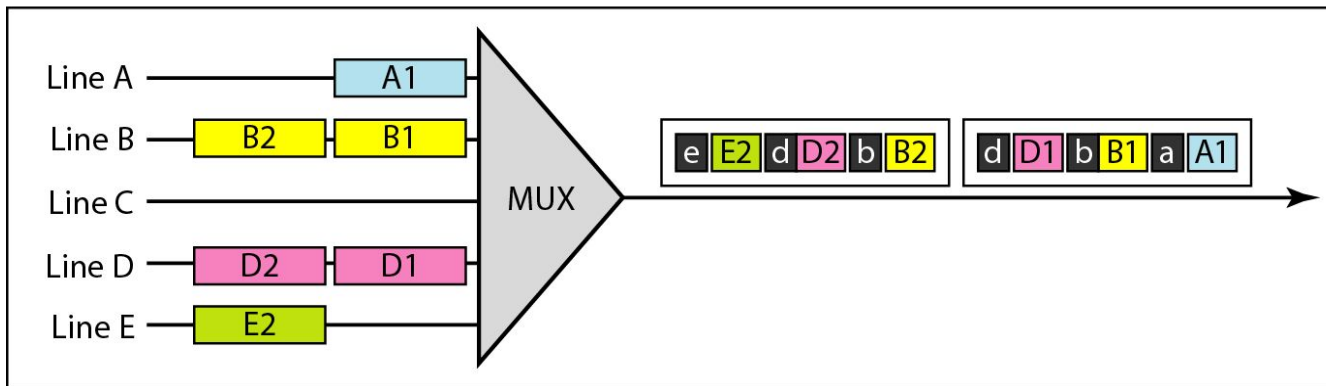
- An application needs to break-up its message in packets, and re-assemble at the receiver
- Fair allocation of link capacity: FIFO or QoS
- Buffer may overflow – *congestion* at the switch



TDM slot comparison



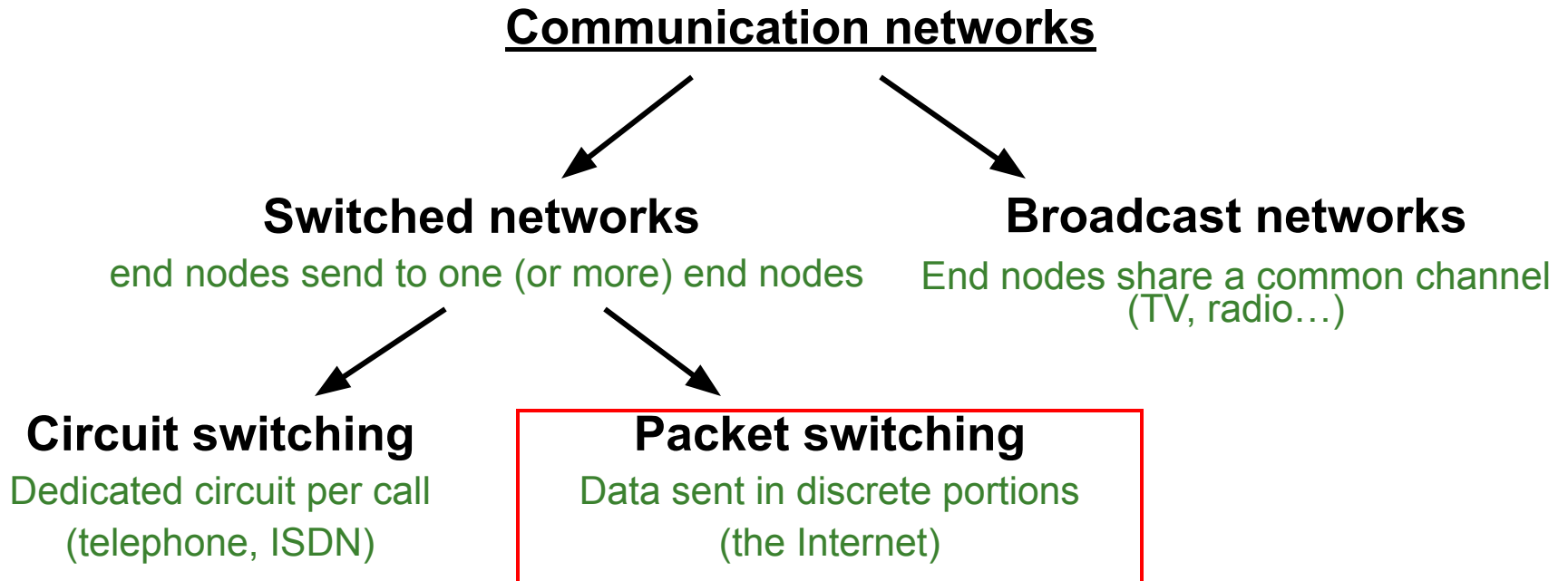
a. Synchronous TDM



b. Statistical TDM

- Slot Size
- No Synchronization Bit
- Bandwidth

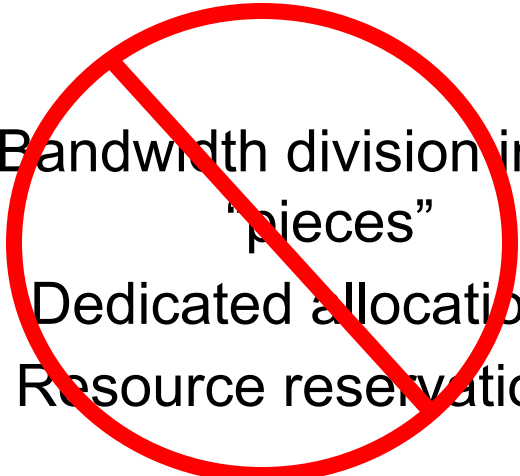
Communication networks



Packet Switching

- each end-end data stream divided into *packets*
- user A, B packets *share* network resources
- each packet uses full link bandwidth
- resources used *as needed*

Bandwidth division into
“pieces”
Dedicated allocation
Resource reservation

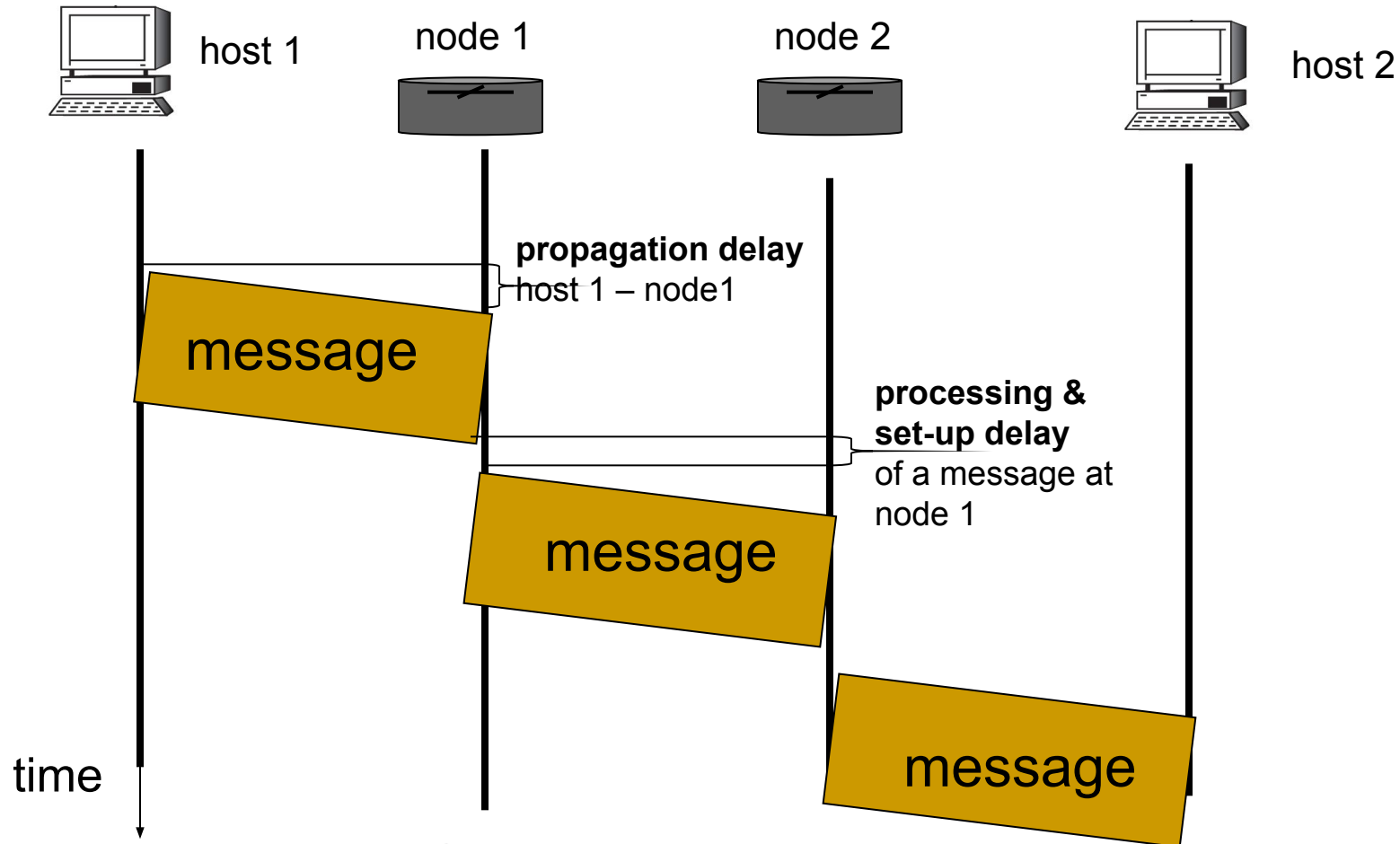


resource contention:

- aggregate resource demand can exceed amount available
- congestion: packets queue, wait for link use
- store and forward:
 - packets move one hop at a time
 - Node receives complete packet before forwarding

Packet switching

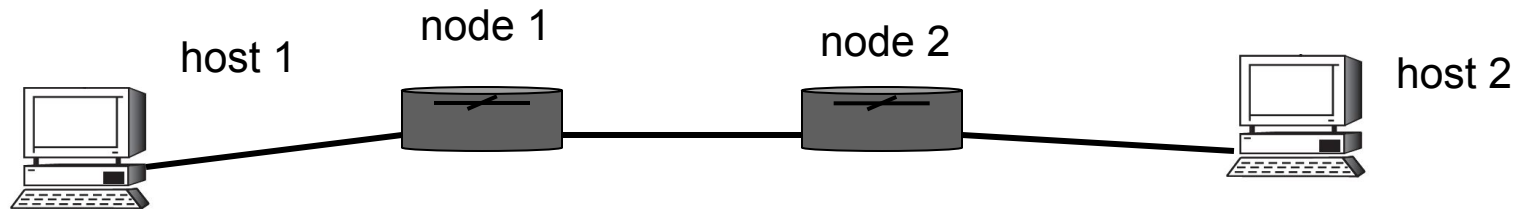
- *Why not message switching?*-



Store-and-Forward

Message switching

EXAMPLE



for simplicity: ignore *processing* and *propagation* delays

$M=7.5$ Mb

transmission delay:

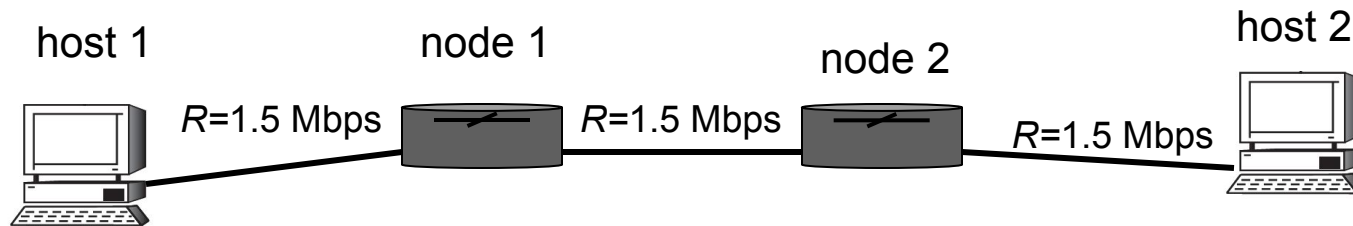
$R=1.5$ Mbps

$$3 \cdot \frac{M}{R} = 15 \text{ [s]}$$

Store complete message and then forward

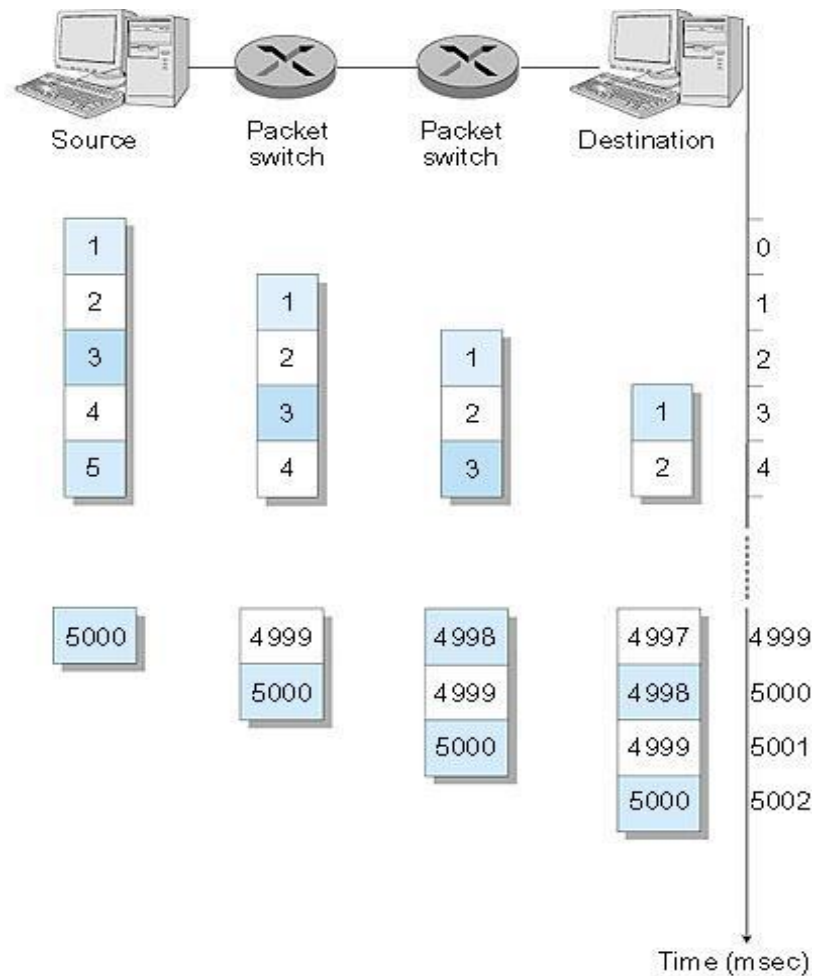
Message switching versus packet switching

- Example

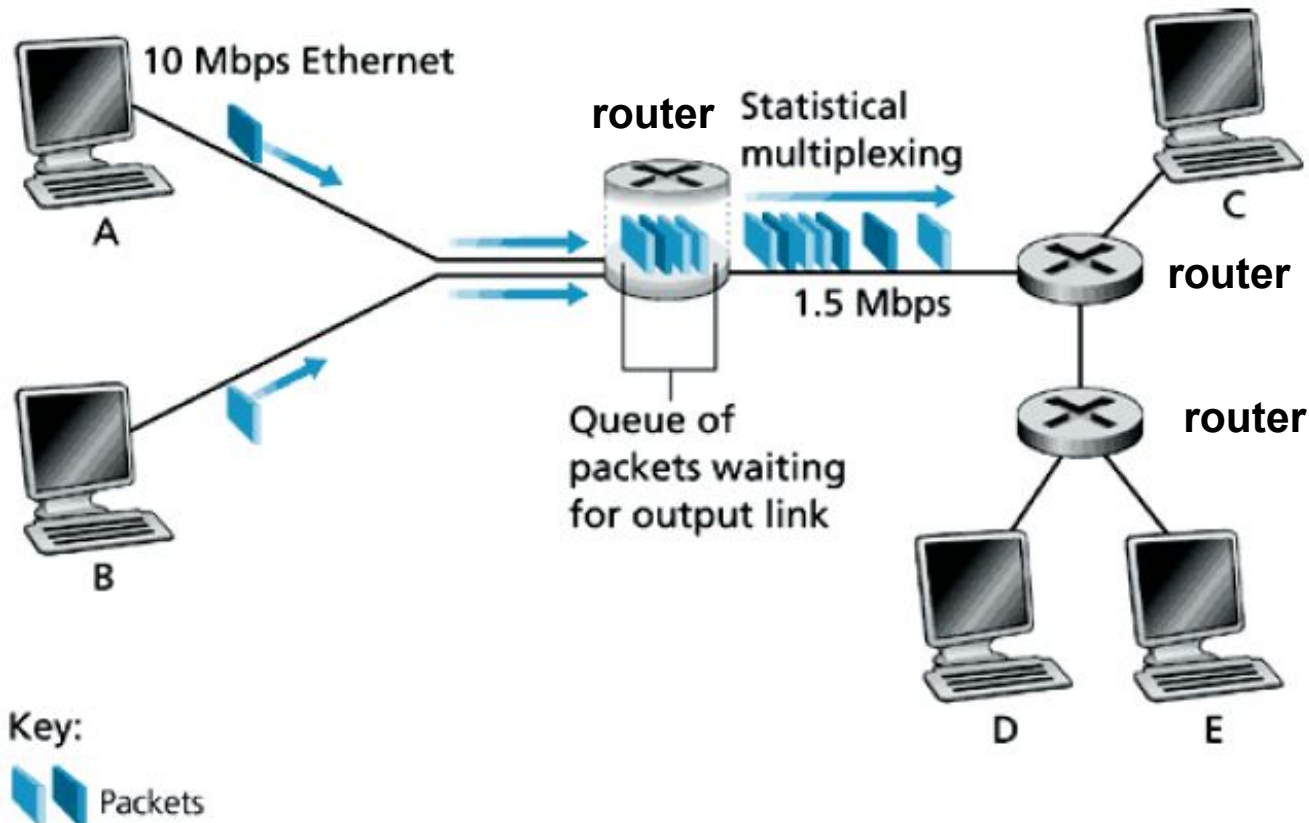


- For simplicity ignore processing and propagation delays
- Split the message into packets each with 1500 bits long
- Store only 1 packet and then forward it
- 1 ms to transmit packet on 1 link
- **Pipelining:** each link works in parallel
- **Delay reduced from 15 s to 5.002 s!!!**

Packet switching



Packet Switching

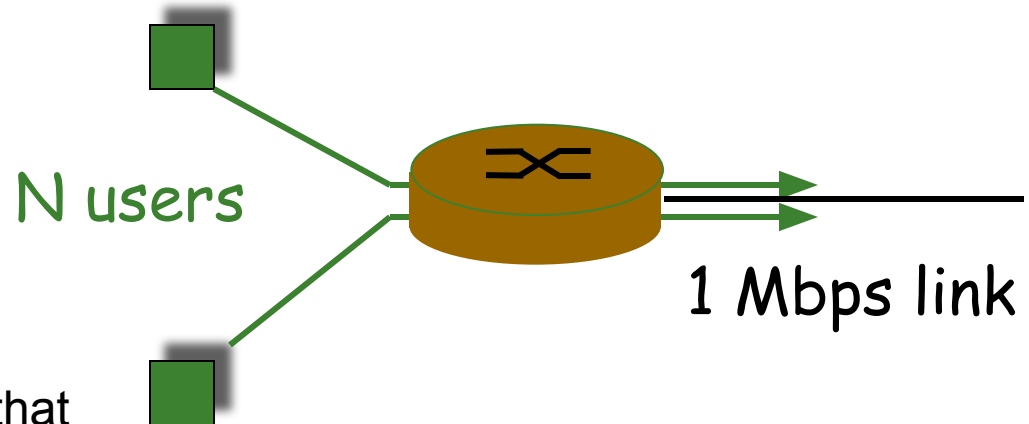


Sequence of A & B packets does not have fixed pattern □ statistical multiplexing.

Packet switching versus circuit switching

Packet switching allows more users to use network!

- 1 Mb/s link
- each user:
 - 100 kb/s when “active”
 - active 10% of time
- circuit-switching:
 - 10 users
- packet switching:
 - with 35 users, probability that there are 11 or more simultaneously active users is approximately .0004



Packet switching versus circuit switching

Is packet switching a “winner?”

- Great for bursty data
 - resource sharing
 - simpler, no call setup
- Excessive congestion: packet delay and loss
 - protocols needed for reliable data transfer, congestion control
- Q: How to provide circuit-like behavior?
 - bandwidth guarantees needed for audio/video apps
 - still an unsolved problem

Packet switching versus circuit switching (cnt'd)

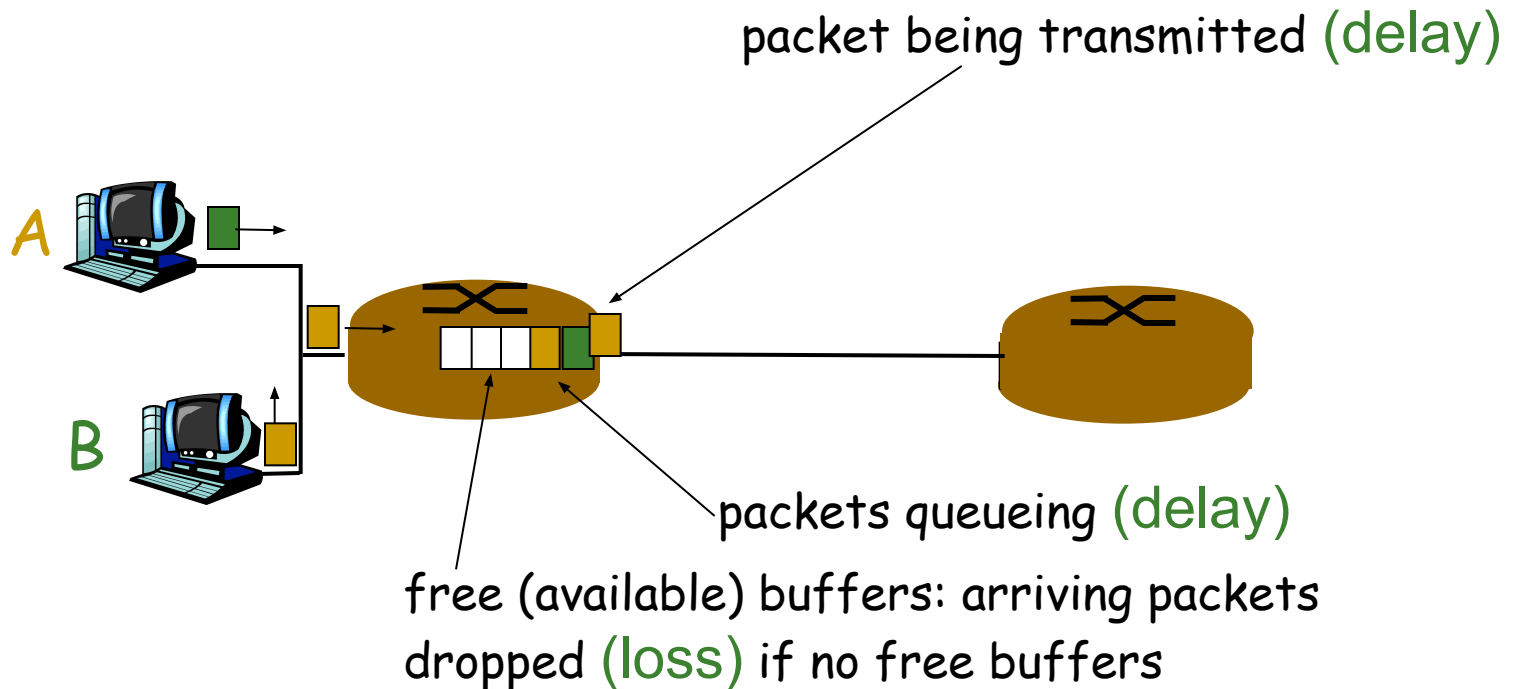
- **Advantages** of packet switching over circuit switching
 - ❑ Statistical multiplexing, and therefore efficient bandwidth usage
 - ❑ Simple to implement

- **Disadvantages** of packet switching over circ. switching
 - ❑ Excessive congestion: packet delay and high loss
 - ❑ Protocols needed for reliable data transfer, congestion control
 - ❑ Packet header overhead
 - ❑ Provides no transparency to a user
 - Analogy: a road versus a railroad

How do loss and delay occur?

packets *queue* in router buffers

- packet arrival rate to link exceeds output link capacity
- packets queue, wait for turn



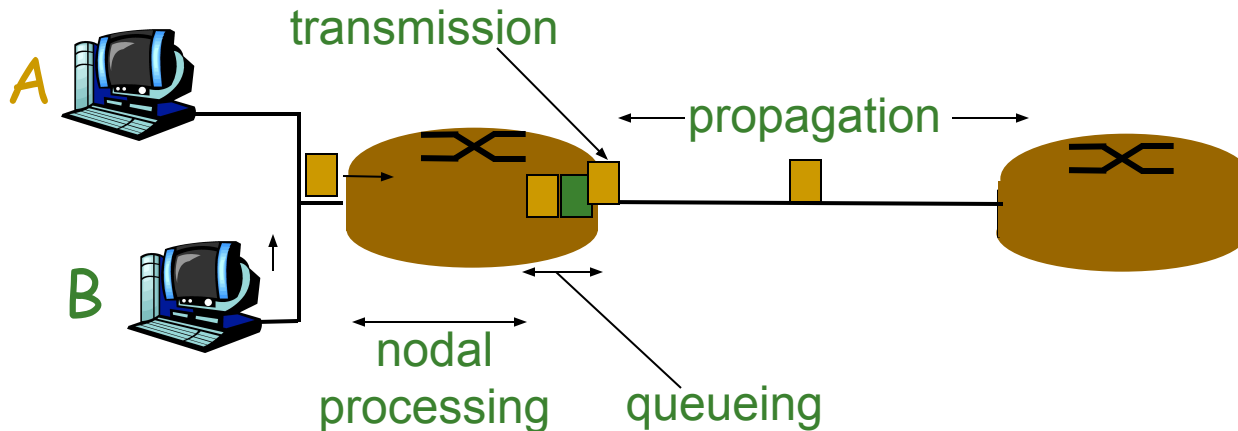
Four sources of packet delay

■ 1. Nodal processing:

- ❑ check bit errors
- ❑ determine output link

■ 2. Queueing

- ❑ time waiting at output link for transmission
- ❑ depends on congestion level of router



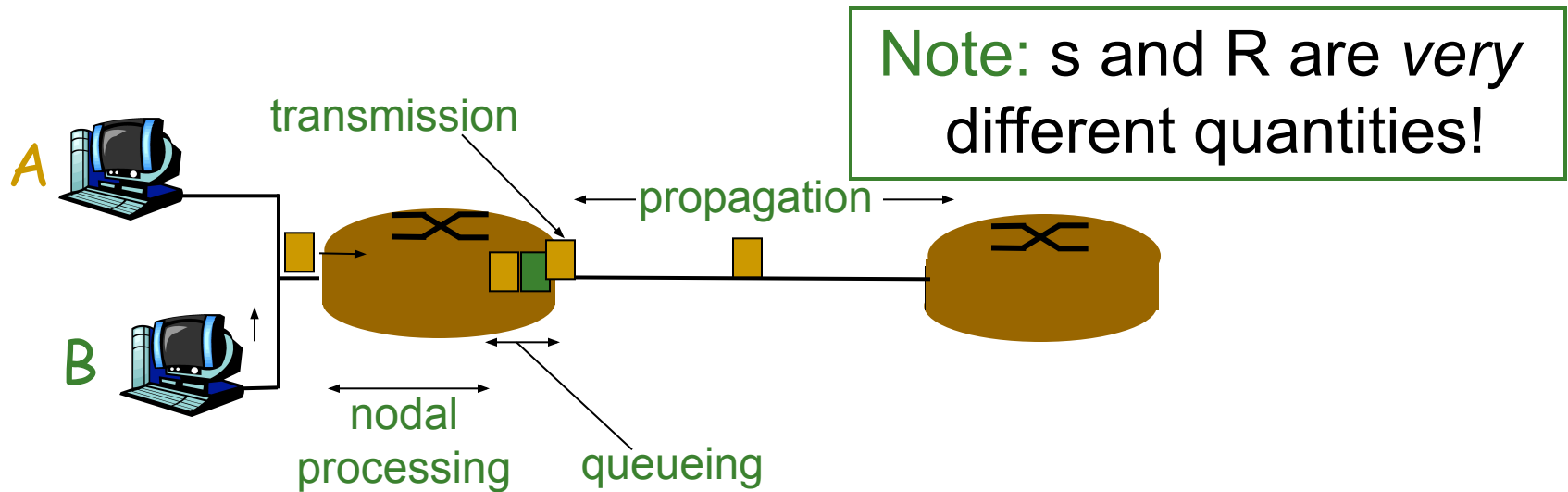
Delay in packet-switched networks

3. Transmission delay:

- R = link bandwidth (bps)
- L = packet length (bits)
- time to send bits into link
= L/R

4. Propagation delay:

- d = length of physical link
- s = propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- propagation delay = d/s



Nodal delay

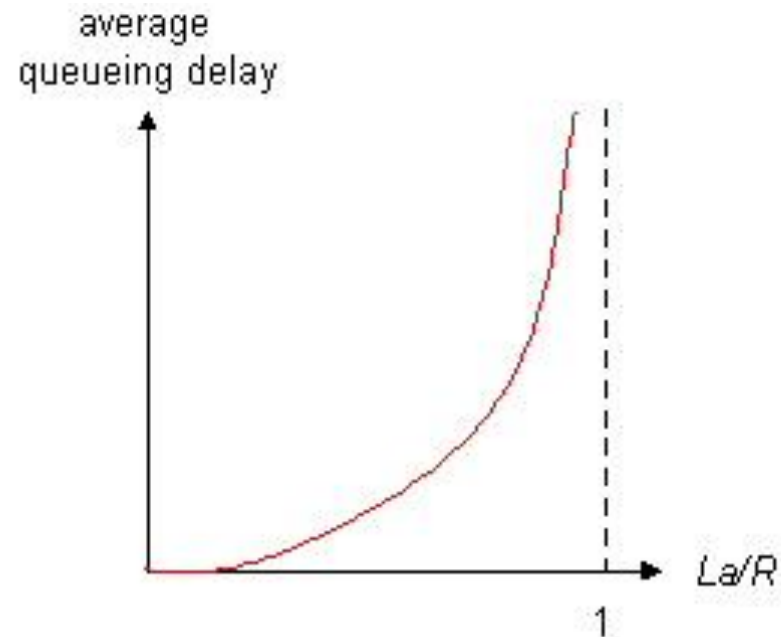
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- d_{proc} = processing delay
 - typically a few microsecs or less
- d_{queue} = queuing delay
 - depends on congestion
- d_{trans} = transmission delay
 - $= L/R$, significant for low-speed links
- d_{prop} = propagation delay
 - a few microsecs to hundreds of msecs

Queueing delay (revisited)

- R =link bandwidth (bps)
- L =packet length (bits)
- a =average packet arrival rate

traffic intensity = La/R



- $La/R \sim 0$: average queueing delay small
- $La/R \rightarrow 1$: delays become large
- $La/R > 1$: more “work” arriving than can be serviced, average delay infinite!

Packet loss

- queue preceding link in buffer has finite capacity
- when packet arrives to full queue, packet is dropped
- lost packet may be retransmitted by previous node, by source end system, or not retransmitted at all

Assignment 1

- You can find Assignment 1 from course web.
- Due Date: First class of Next Week

Quiz 1

- On the day of submission of Assignment related with topics covered in Assignment 1.

Readings

- Computer Networking, a top-down approach featuring the Internet (3rd edition), J.K.Kurose, K.W.Ross
 - Chapter 1: Section 1.3, 1.6

